Constraint Satisfaction

CS5491: Artificial Intelligence

ZHICHAO LU

Content Credits: Prof. Wei's CS4486 Course
and Prof. Boddeti's AI Course

TODAY

Part 1: Constraint Satisfaction Problems

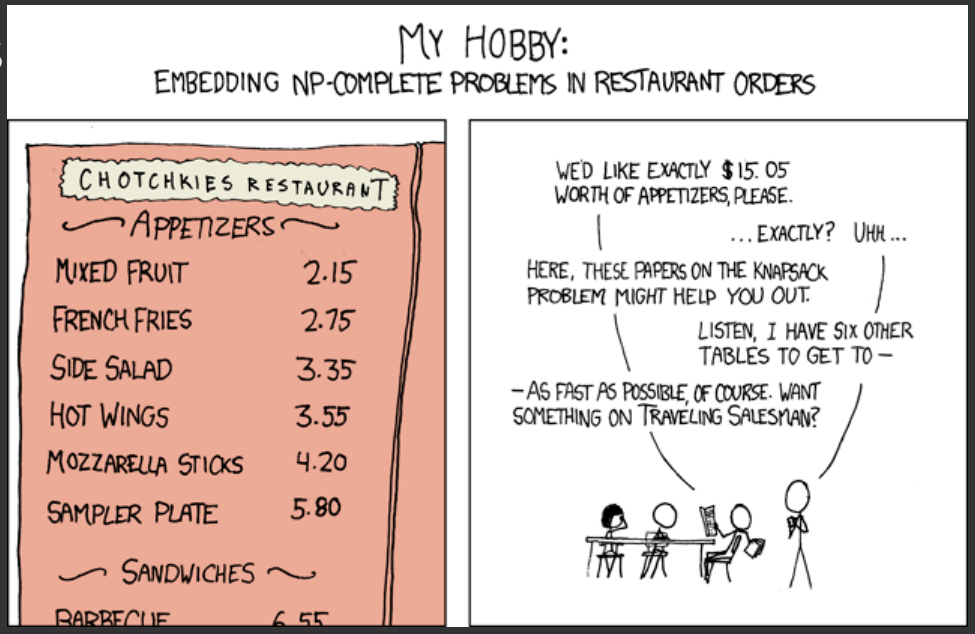
- Formulation, examples, varieties
- Naive search over CSPs

Part 2: Backtracking Search

- Filtering, ordering, structure

Reading

- Today's Lecture: RN Chapter 6
- Next Lecture: RN Chapter 5



XKCD

WHAT IS SEARCH FOR?



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Given assumptions about the world: a single agent, deterministic actions, fully observed state, discrete state space



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  - Paths have various costs, depths
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- Planning: sequences of actions
  - The path to the goal is the important thing
  - Paths have various costs, depths
  - Heuristics give problem-specific guidance
- Identification: assignments to variables
  - The goal itself is important, not the path
  - All paths at the same depth (for some formulations)
  - CSPs are a specialized class of identification problems



CONSTRAINT SATISFACTION PROBLEMS



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Standard search problems:

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- Goal test can be any function over states
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Constraint satisfaction problems (CSPs):

- A special subset of search problems
- State is defined by variables $\mathbf{X}_i$ with values from a domain D (sometimes D depends on i)
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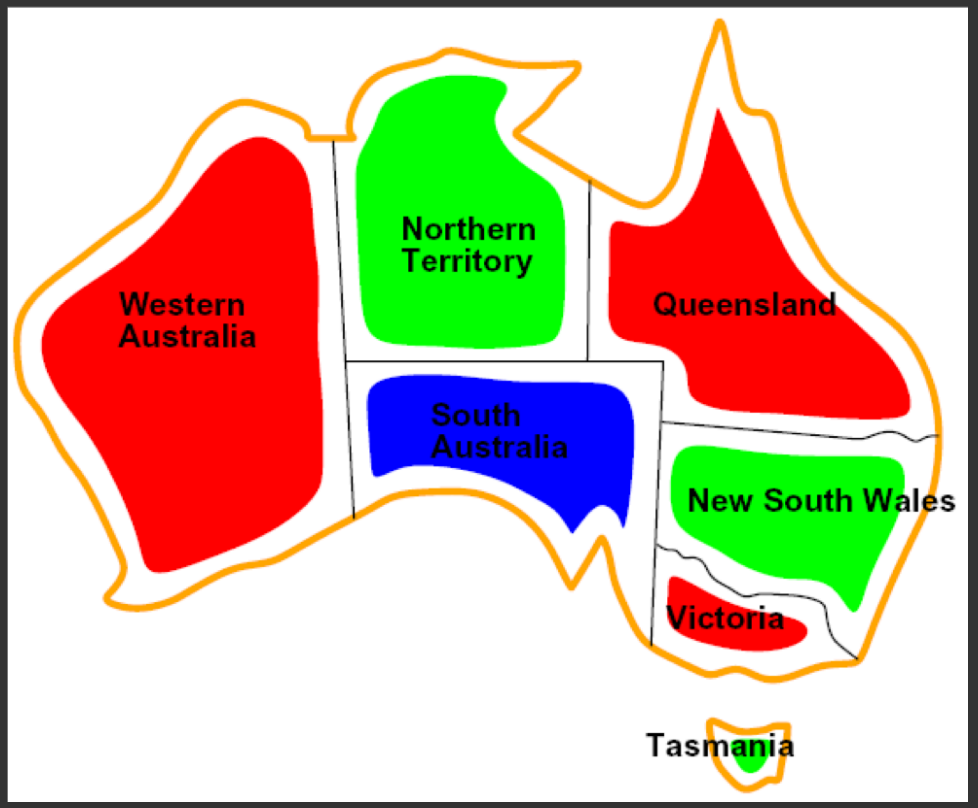
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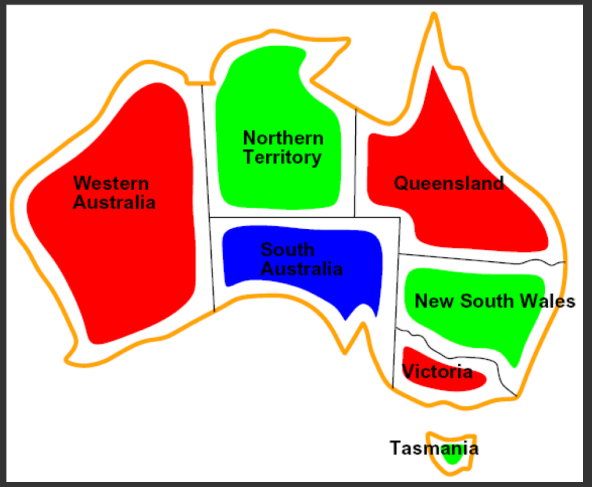
Allows useful general-purpose algorithms with more power than standard search algorithms



CSP EXAMPLE

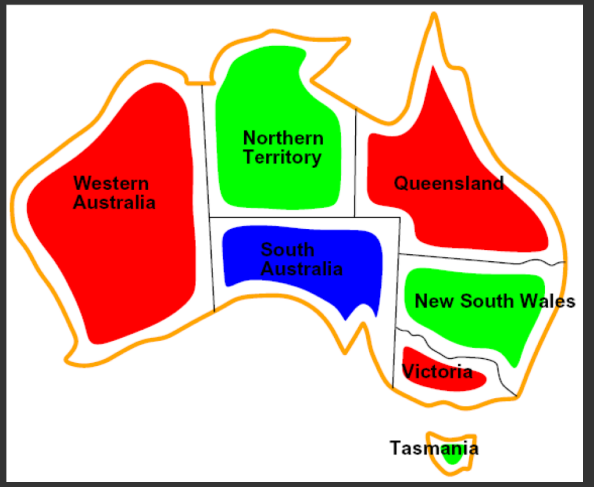

Western Australia
Northern Territory
Queensland
South Australia
New South Wales
Victoria
Tasmania

EXAMPLE: MAP COLORING



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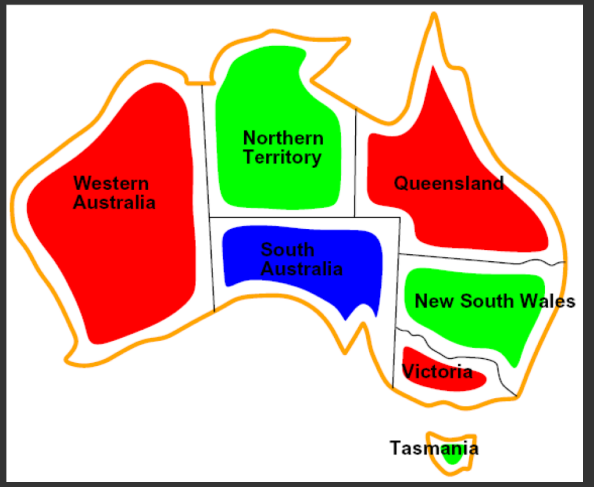
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Domains: $D = \{red, green, blue\}$

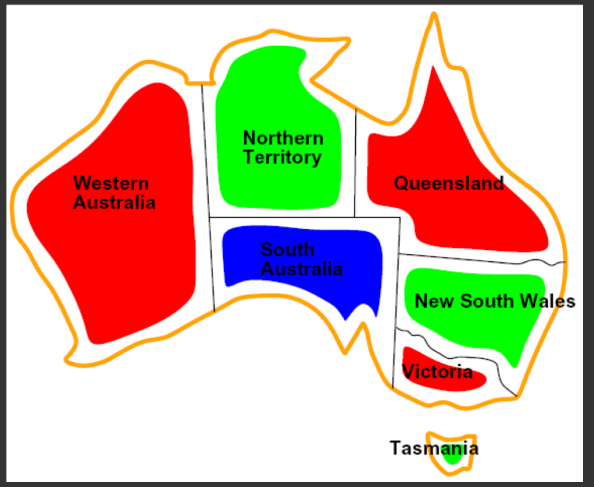


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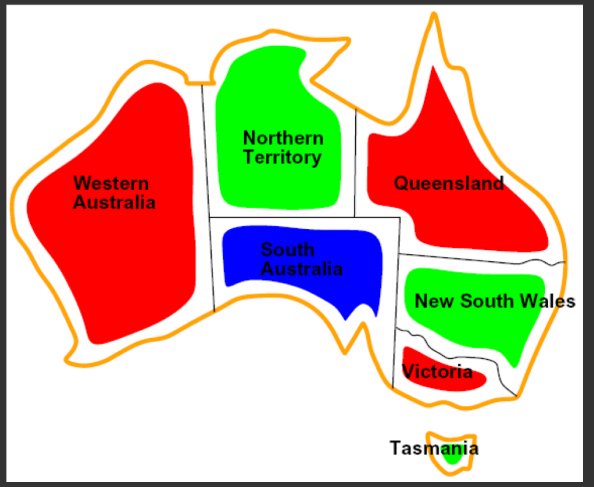
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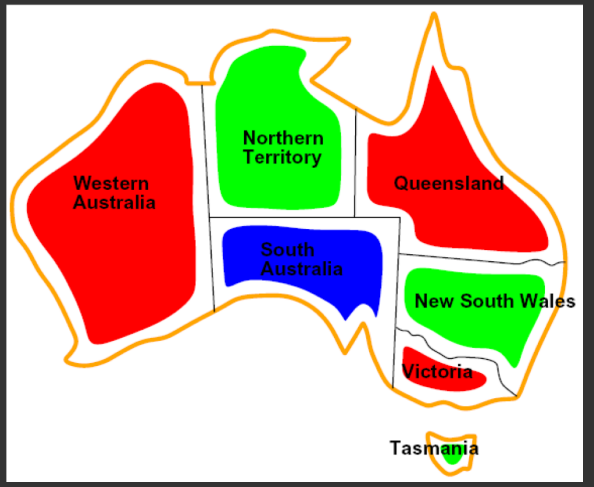
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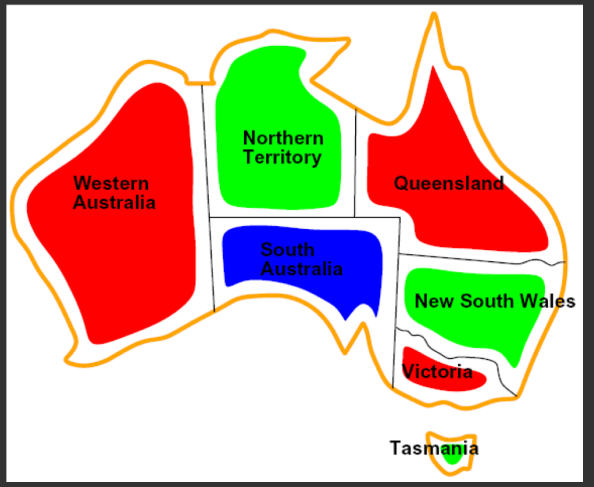
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Solutions: Assignments that satisfy all constraints, e.g.:

- {WA=red, NT=green, Q=red, NSW=green, V=red, SA=blue, T=green}



CONSTRAINT GRAPHS



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Binary CSP: each constraint relates two variables



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Constraint Graph: nodes are variables, arcs are constraints



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General-purpose CSP algorithms use the graph structure to speed up search. E.g., Tasmania is an independent subproblem.

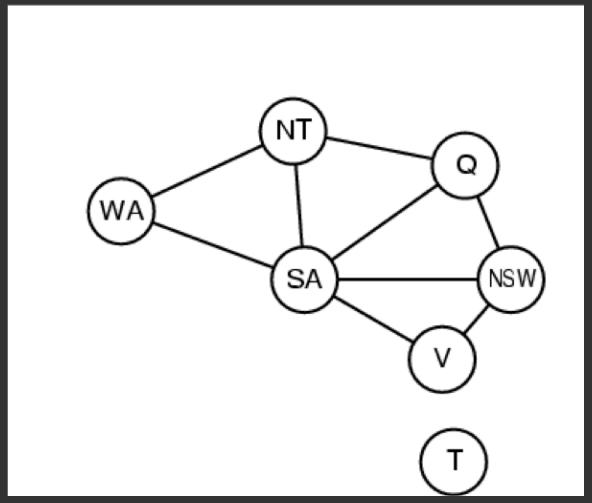


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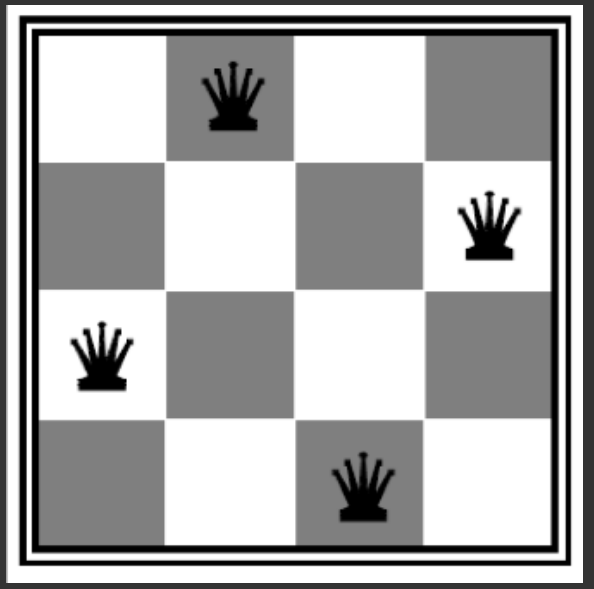
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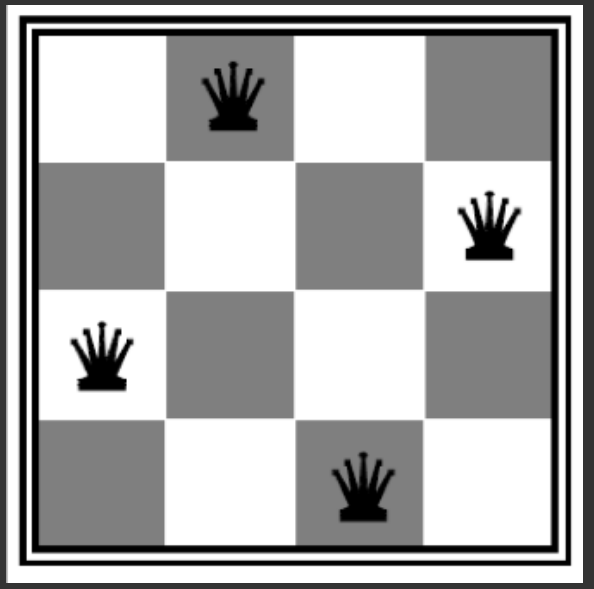


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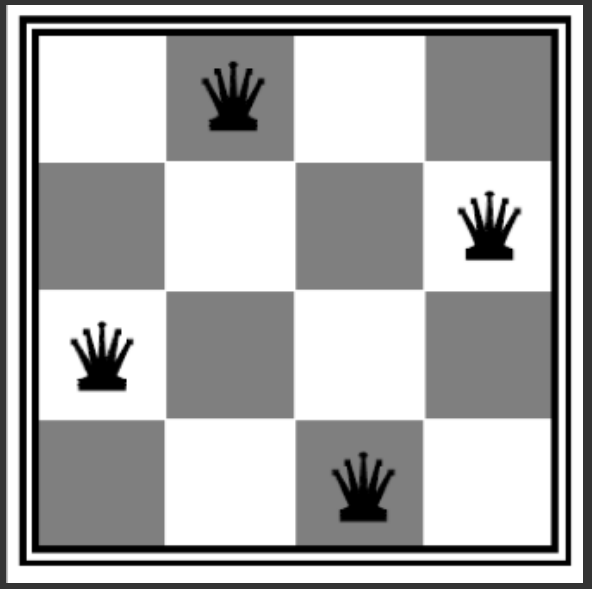
Formulation 1:



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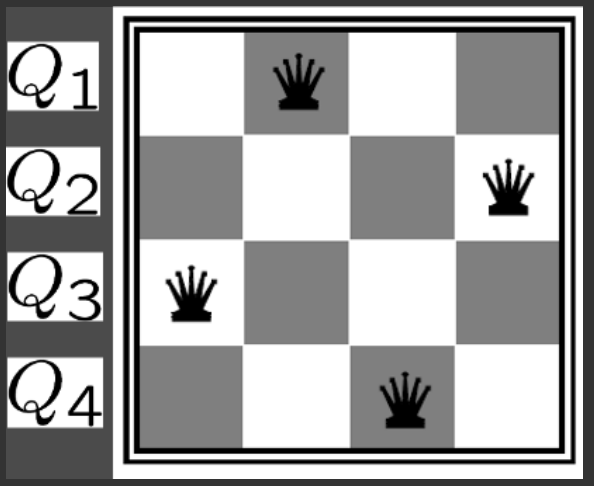
Formulation 1:

- Variables: X_{ij}
- Domains: $\{0, 1\}$
- Constraints:



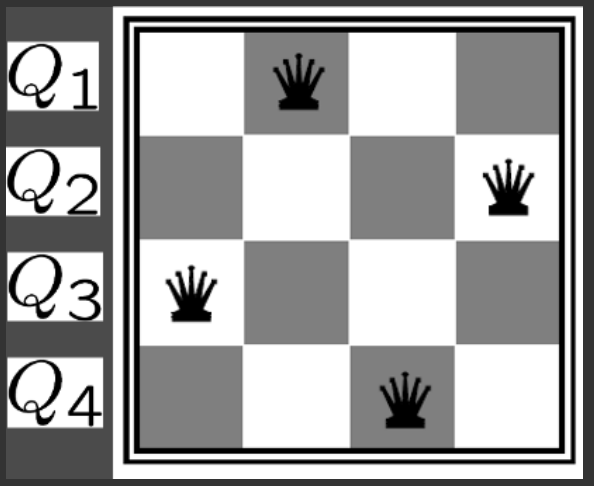
$$\forall\, i, j, k \;\; (X_{ij}, X_{ik}) \in \{(0,0), (0,1), (1,0)\}$$
$$\forall\, i, j, k \;\; (X_{ij}, X_{kj}) \in \{(0,0), (0,1), (1,0)\}$$
$$\forall\, i, j, k \;\; (X_{ij}, X_{i+k,j+k}) \in \{(0,0), (0,1), (1,0)\}$$
$$\forall\, i, j, k \;\; (X_{ij}, X_{i+k,j-k}) \in \{(0,0), (0,1), (1,0)\}$$
$$\sum_{ij} X_{ij} = N$$

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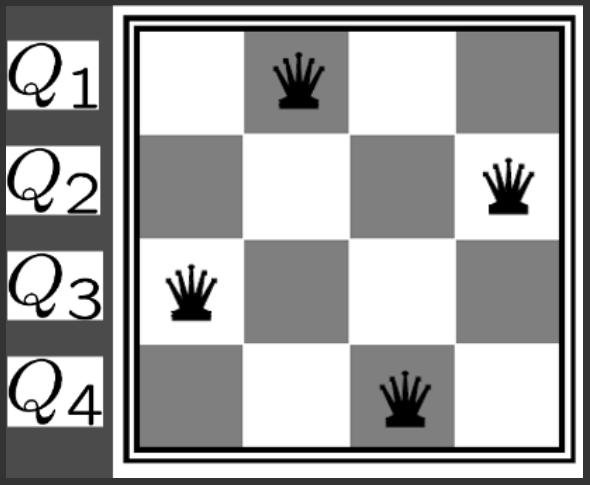
Formulation 2:



EXAMPLE: N QUEENS

Formulation 2:

- Variables: Q_k
- Domains: $\{1, 2, 3, \dots, N\}$
- Constraints:
  - Implicit: $\forall (i, j)$ $\textbf{non-threatening}$ (Q_i, Q_j)
  - Explicit: $(Q_1, Q_2) \in \{(1, 3), (1, 4), \dots\}$



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Continuous variables

- E.g., start/end times for Hubble Telescope observations
- Linear constraints solvable in polynomial time by LP methods



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Preferences (soft constraints):

- E.g., red is better than green
- Often representable by a cost for each variable assignment
- Leads to constrained optimization problems



REAL-WORLD CSPS

Scheduling problems: e.g., when can we all meet?

Timetabling problems: e.g., which class is offered when and where?

Assignment problems: e.g., who teaches what class

Hardware configuration

Transportation scheduling

Factory scheduling

Circuit layout

Fault diagnosis

. . .



SOLVING CSPS



STANDARD SEARCH FORMULATION

Standard search formulation of CSPs



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States defined by the values assigned so far (partial assignments)

- Initial state: the empty assignment, {}
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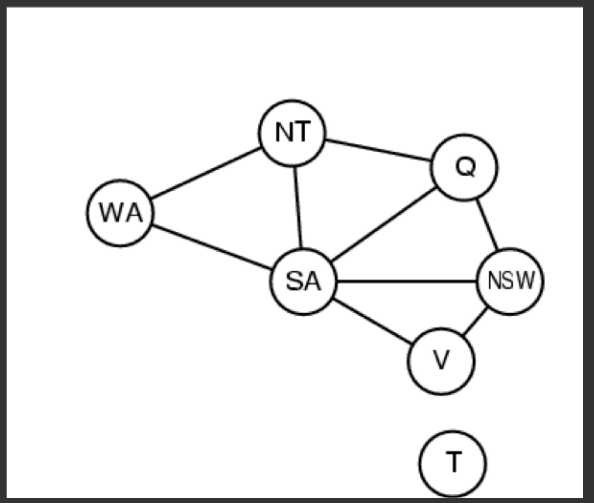
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Start with straightforward search and then improve.



SEARCH METHODS

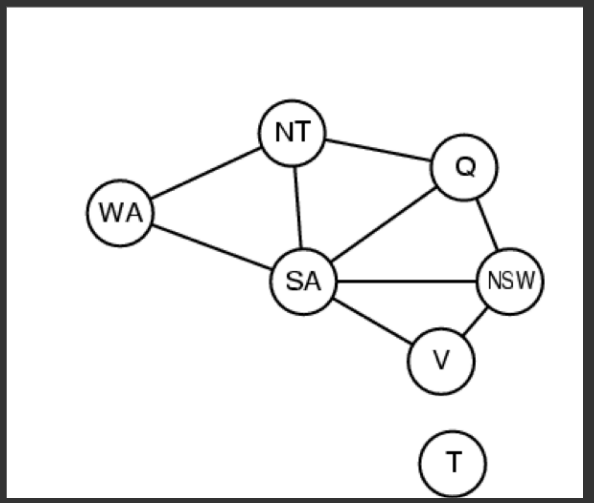
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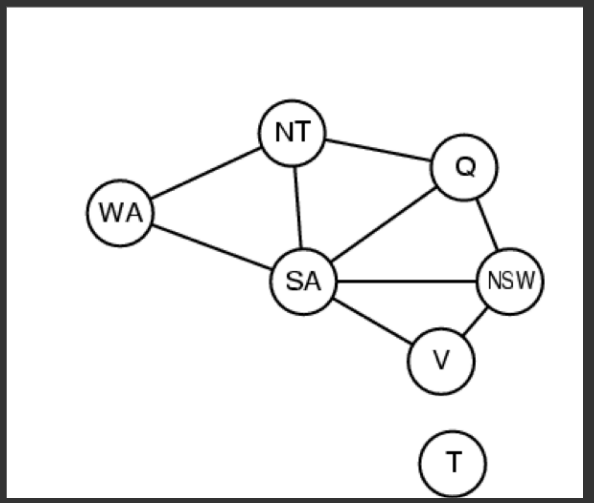


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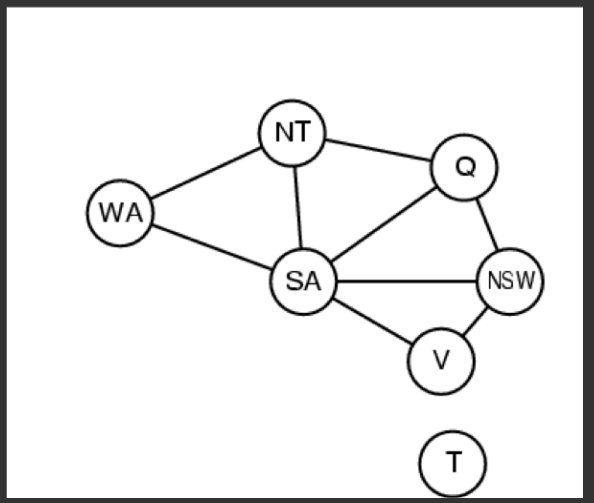
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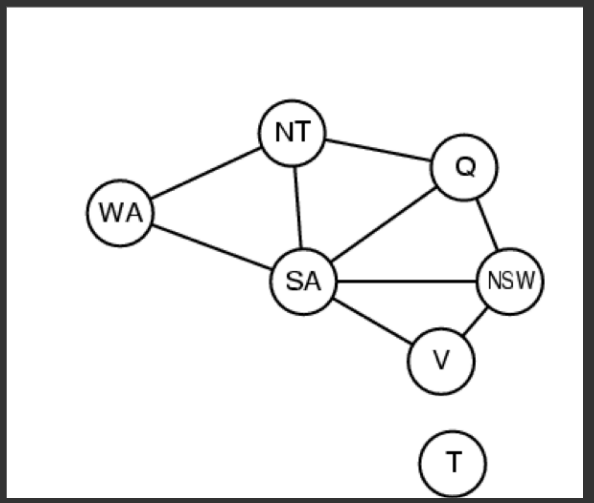
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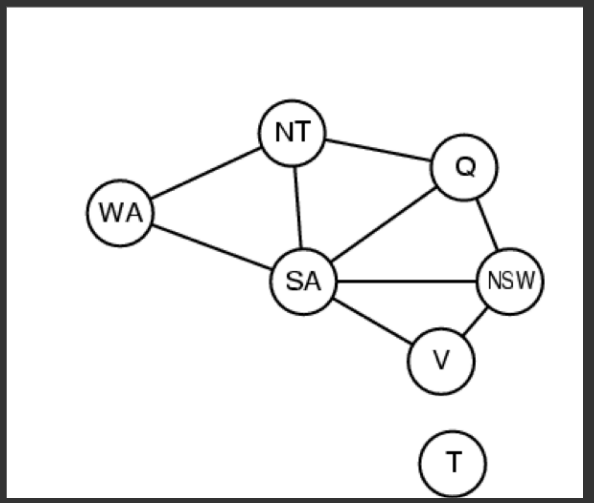
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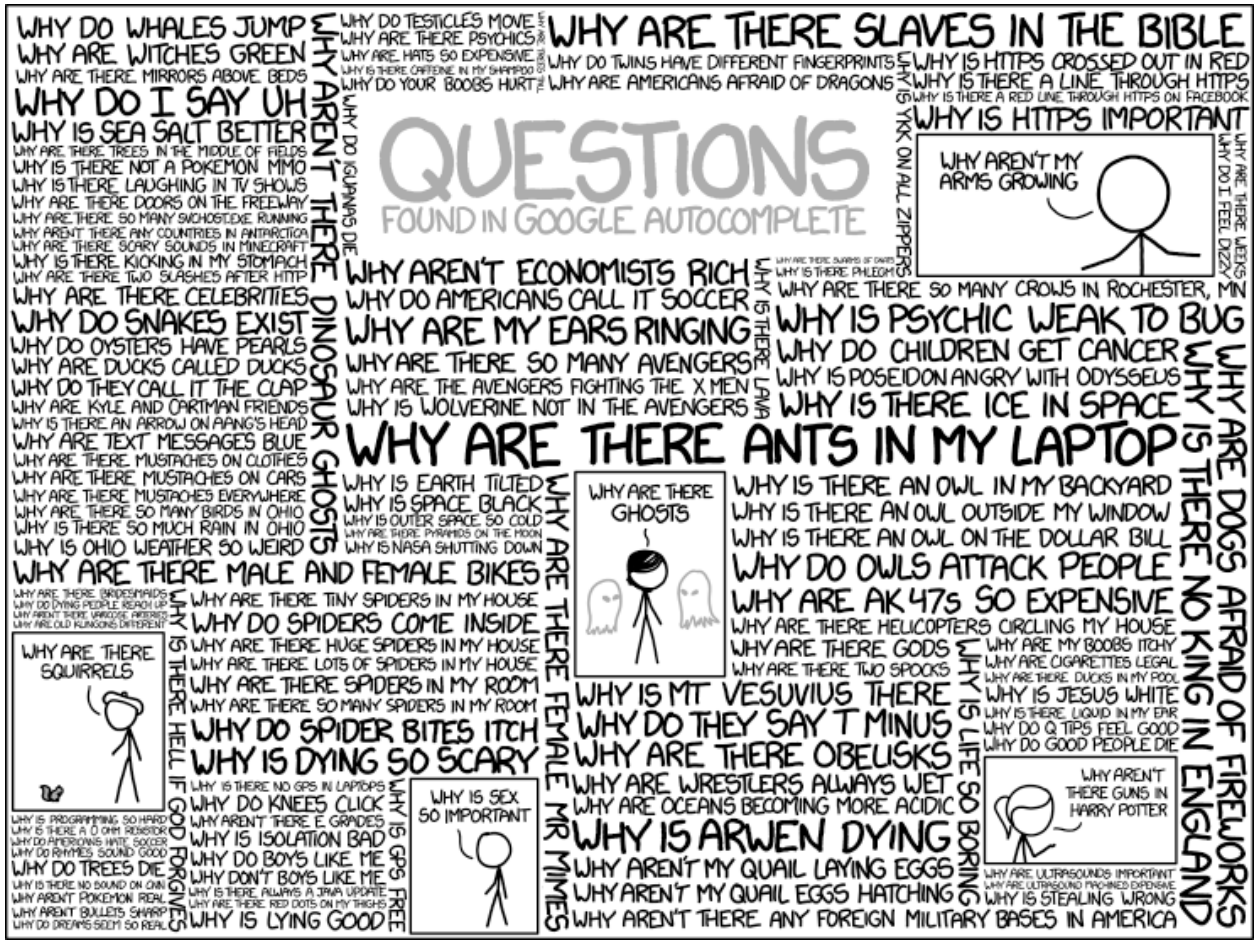
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What problems does naïve search have?

- Ignores that assignments commute ($n!$ orderings of the same assignment); detects violations only at the end



Q & A


QUESTIONS
FOUND IN GOOGLE AUTOCOMPLETE
WHY ARE THERE SLAVES IN THE BIBLE
WHY ARE THERE ANTS IN MY LAPTOP
WHY ARE THERE GHOSTS
WHY ARE THERE SQUIRRELS
WHY IS SEX SO IMPORTANT
WHY AREN'T MY ARMS GROWING
WHY AREN'T THERE GUNS IN HARRY POTTER
WHY DO WHALES JUMP
WHY ARE WITCHES GREEN
WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SWOOSH.EXE RUNNING
WHY AREN'T THERE ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY ARE THERE MUSTACHES ON CARS
WHY ARE THERE MUSTACHES EVERYWHERE
WHY ARE THERE SO MANY BIRDS IN OHIO
WHY IS THERE SO MUCH RAIN IN OHIO
WHY IS OHIO WEATHER SO WEIRD
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REACH UP
WHY AREN'T THERE VARICOSE ARTERIES
WHY ARE OLD KUNGONS DIFFERENT
WHY IS PROGRAMMING SO HARD
WHY IS THERE A 0 OHM RESISTOR
WHY DO AMERICANS HATE SOCCER
WHY DO RHYMES SOUND GOOD
WHY DO TREES DIE
WHY IS THERE NO SOUND ON OVA
WHY AREN'T POKEMON REAL
WHY AREN'T BULLETS SHARP
WHY DO DREAMS SEEM SO REAL
WHY DO IGUANAS DIE
WHY DO TESTICLES MOVE
WHY ARE THERE PSYCHICS
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY DO YOUR BOOBS HURT
WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE AMERICANS AFRAID OF DRAGONS
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY ARE THERE SO MANY AVENGERS
WHY ARE THE AVENGERS FIGHTING THE X MEN
WHY IS WOLVERINE NOT IN THE AVENGERS
WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPIDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY AREN'T THERE E GRADES
WHY IS ISOLATION BAD
WHY DO BOYS LIKE ME
WHY DON'T BOYS LIKE ME
WHY IS THERE ALWAYS A JAVA UPDATE
WHY ARE THERE RED DOTS ON MY THIGHS
WHY IS LYING GOOD
WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE AK 47s SO EXPENSIVE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOOKS
WHY IS MT VESUVIUS THERE
WHY DO THEY SAY T MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMING MORE ACIDIC
WHY IS ARWEN DYING
WHY AREN'T MY QUAIL LAYING EGGS
WHY AREN'T MY QUAIL EGGS HATCHING
WHY AREN'T THERE ANY FOREIGN MILITARY BASES IN AMERICA
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO GOOD PEOPLE DIE
WHY ARE ULTRASOUNDS IMPORTANT
WHY ARE ULTRASOUND MACHINES EXPENSIVE
WHY IS STEALING WRONG
WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON FACEBOOK
WHY IS HTTPS IMPORTANT
WHY IS YKK ON ALL ZIPPERS
WHY ARE THERE SWARMS OF GNATS
WHY IS THERE PHLEGM
WHY ARE THERE SO MANY CROWS IN ROCHESTER, MN
WHY IS PSYCHIC WEAK TO BUG
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY ARE DOGS AFRAID OF FIREWORKS
WHY IS THERE NO KING IN ENGLAND
WHY ARE THERE FEMALE MR MIMES
WHY IS THERE HELL IF GOD FORGIVES
WHY IS GPS FREE
WHY IS LIFE SO BORING
WHY ARE THERE LEEKS
WHY DO I FEEL DIZZY

XKCD